

Name Class Date **P4B - Chapter 3 quiz**

Total questions: 15

Worksheet time: 22mins

1. You want to create a dataset in **SAS** that stores information about several flights, including the **Flight ID, Airline, Origin, Destination, number of Passengers, and Delay time**, using raw data inputation.

Which of the following SAS code snippets correctly creates the dataset?

a)

```
DATA flights;
  INPUT FlightID $ Airline $ Origin $ Destination $ Passengers Delay;
  DATALINES;
  F101 Delta JFK LAX 100 15
  F102 United LAX ORD 150 0
  F103 Lufthansa FRA JFK 220 30
  F104 TAP LIS MAD 90 5
  ;
RUN;
```

b)

```
DATA flights;
  INPUT FlightID Airline $ Origin $ Destination $ Passengers Delay;
  DATALINES;
  F101 Delta JFK LAX 100 15
  F102 United LAX ORD 150 0
  F103 Lufthansa FRA JFK 220 30
  F104 TAP LIS MAD 90 5
  ;
RUN;
```

c)

```
DATA flights;
  INPUT FlightID $ Airline $ Origin $ Destination $ Passengers Delay;
  CARDS;
  F101 Delta JFK LAX 100 15
  F102 United LAX ORD 150 0
  F103 Lufthansa FRA JFK 220 30
  F104 TAP LIS MAD 90 5;
RUN;
```

d)

```
DATA flights;
  INPUT FlightID $ Air line $ Origin $ Destination $ Passengers Delay;
  DATALINES;
  F101 Delta JFK LAX 100 15
  F102 United LAX ORD 150 0
  F103 Lufthansa FRA JFK 220 30
  F104 TAP LIS MAD 90 5
  ;
RUN;
```

2.

```
DATA flights;
SET flights;
RUN;
```

The code shown in the image was executed in SAS. What does this code do?

- a) It creates a new dataset called flights. b) It overwrites the already existing dataset flights.
- c) The code does not run, it contains an error. d) None of the above.
3.

```
DATA ch3res.churn2;
  SET ch3res.churn;
  WHERE SURNAME LIKE 'Abe_';
RUN;
```

Using the dataset 'churn' from the classes and based on the code below, what surname from the ones below can be retrieved from this query?

- a) Abigal b) Abeer
- c) Abesh d) Abel

```
4. DATA students;  
   INPUT StudentID $ Name $ Score;  
   DATALINES;  
   S01 Ana 12  
   S02 Bruno 8  
   S03 Carla 14  
   S04 Diego 18  
   ;  
   RUN;
```

Consider the code in the image to create an SAS dataset containing information about students and their exam scores.

The objective is to create a **new variable called Passed** that stores **1** if the **student's score is 10 or higher** and **0** otherwise.

Which of the following is correct?

- | | |
|---|--|
| a) DATA students_new;
SET students;
Passed = Score >= 10 THEN 1 ELSE 0;
RUN; | b) DATA students_new;
SET students;
Passed == 1 IF Score >= 10;
RUN; |
| c) DATA students_new;
SET students;
IF Score >= 10 THEN Passed = 1;
ELSE Passed = 0;
RUN; | d) DATA students_new;
SET students;
IF Score >= 10 Passed = 1;
ELSE Passed = 0;
RUN; |

5. The WHERE statement can only be used on DATA steps.

- a) True b) False

6. The DROP and KEEP options can be applied in the DATA statement, the SET statement, or used as standalone statements within a DATA step to control which variables are included or excluded?

- a) True b) False

```
7. DATA employees_new;  
SET ch3res.employees;  
LENGTH SalaryLevel $4;  
SELECT;  
    WHEN (Salary < 30000) SalaryLevel = 'Low';  
    WHEN (Salary < 70000 AND Salary > 30000) SalaryLevel = 'Medium';  
    OTHERWISE SalaryLevel = 'High';  
END;  
RUN;
```

Based on the code in the image, if an employee earns 60000, what is the value in the SalaryLevel variable?

- a) Medium b) Low
- c) High d) None of the above

8.

```
DATA ch3res.smsales2 (Keep=Invoice_ID Product_Line Tax_5 Total Date Total_Without_Tax);  
SET ch3res.smsales;  
Total_Without_Tax = Total - Tax_5;  
WHERE Product_Line LIKE 'Food and beverages';  
LABEL Total = 'Total price including tax'  
      Total_Without_Tax = 'Total price excluding tax';  
RUN;
```

After executing the following SAS code, we want to filter observations where the variable labeled *'Total price including tax'* is greater than 50.

Can we use the label *'Total price including tax'* directly in the IF statement, as shown below?

```
DATA smsales;  
SET ch3res.smsales2;  
IF 'Total price including tax' > 50;  
RUN;
```

a) True

b) False

9. A student wants to compute a **cumulative total salary** using the following code:

```
DATA employees_new;  
SET employees;  
RETAIN Total;  
Total = Total + Salary;  
RUN;
```

Knowing that the variable Salary possesses missing values, what will happen to the variable Total?

- | | |
|---|---|
| a) The cumulative total is correctly calculated for all observations. | b) The cumulative total becomes missing when a missing Salary is encountered and remains missing afterward. |
| c) The cumulative total skips missing values and continues correctly. | d) The code produces an error due to missing values. |

```
10. DATA salaries;  
    INPUT Salary;  
    DATALINES;  
1000  
2000  
.  
1500  
1200  
;  
RUN;
```

```
DATA salaries;  
    SET salaries;  
    Total + Salary;  
RUN;
```

A student created the code presented in the image.

What will be the value of Total at **line 3** (when Salary = .)?

- a) 1000 b) .
- c) 3000 d) '3000'

- Using the churn dataset you want to create a new variable called **Valuable_Customer** that follows the following rule:

1 → if the customer **is active AND** has a credit card

 $0 \rightarrow \text{otherwise}$

Which of the following SAS code snippets correctly creates the variable? (you can choose more than 1)

a)

```
DATA churn;
  SET ch3res.churn;
  IF IsActiveMember = 1 AND HasCrCard = 1 THEN Valuable_Customer = 1;
  ELSE Valuable_Customer = 0;
RUN;
```

```
b) DATA churn;  
    SET ch3res.churn;  
    IF IsActiveMember = 1 THEN DO;  
        IF HasCrCard = 1 THEN Valuable_Customer = 1;  
        ELSE Valuable_Customer = 0;  
    END;  
    ELSE Valuable_Customer = 0;  
RUN;
```

```
DATA churn;  
  SET ch3res.churn;  
  IF IsActiveMember + HasCrCard = 2 THEN Valuable_Customer = 1;  
  ELSE Valuable_Customer = 0;  
RUN;
```

```
d) DATA churn;  
    SET ch3res.churn;  
    Valuable_Customer = IsActiveMember * HasCrCard;  
RUN;
```

```
12. DATA sales_bonus;  
SET sales;  
    SELECT (Region);  
        WHEN ("North") Bonus = 10;  
        WHEN ("South") Bonus = 15;  
        WHEN ("East") Bonus = 12;  
        OTHERWISE Bonus = 5;  
    END;  
RUN;
```

Based on the code in the image, classify the following statement:

"The SELECT statement will produce an error because Region is not specified in the OTHERWISE statement."

- a) True b) False

13. If we don't put the END statement in a select statement, the code will run smoothly.

- a) True b) False

```
14. DATA churn1;  
SET ch3res.churn (KEEP=CustomerID EstimatedSalary);  
WHERE Geography = 'France';  
RUN;
```

Based on the code in the image, what will be the result returned after running the code?

- a) A syntax error.
- b) A temporary dataset with 2 variables with transactions belonging to France.
- c) A permanent dataset with 2 variables with transactions belonging to France.
- d) An error because Geography is not in the dataset.

```

15.      1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
        68
        69      DATA churn1;
        70      SET ch3res.churn;
        71      RUN
        72
        73      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
          22
          76
ERROR 22-322: Syntax error, expecting one of the following: ;, CANCEL, PGM.
ERROR 76-322: Syntax error, statement will be ignored.

```

After running some lines of code, after checking the log the following error was found. What can be a plausible explanation for it?

- a) A variable in the SET statement does not exist.
- b) The dataset referenced in the SET statement is empty.
- c) A semicolon is missing at the end of a statement.
- d) The WHERE condition filtered out all observations.

Answer Keys

1. a)
2. b)
3. d)
4. c)
5. b)
6. a)
7. d)
8. b)
9. b)
10. c)
11. a), b), c), d)
12. b)
13. b)
14. d)
15. c)